

BREWSTER FIELD, NE, USAWMO: **725628**Lat: **40.450N**Lon: **99.339W**Elev: **2304**StdP: **13.51**Time Zone: **-6.00 (NAC)**Period: **96-19**WBAN: **04935****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.2	4.8	-8.3	3.8	2.0	-3.3	5.0	6.3	30.4	31.0	27.1	31.3	9.0	350	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	95.0	71.8	91.3	71.1	89.5	71.2	76.3	87.6	74.7	86.0	73.2	84.5	12.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.9	133.0	82.3	71.8	128.0	81.2	70.0	120.1	79.6	41.4	87.5	39.9	85.4	38.4	84.9	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.2	25.1	22.2	-7.2	100.3	3.3	2.9	-9.6	102.4	-11.6	104.1	-13.4	105.7	-15.8	107.8		
(5)			-7.5	79.6	3.7	1.4	-10.2	80.7	-12.4	81.5	-14.5	82.3	-17.2	83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	28.2	30.2	40.7	50.2	60.8	71.3	76.0	73.1	65.8	52.5	40.6	29.5		
	DBStd	19.36	11.54	11.65	11.43	9.62	8.68	6.74	5.51	5.51	8.17	9.47	9.70	10.64		
	HDD50	2707	677	558	324	112	13	0	0	0	2	81	305	635		
	HDD65	5853	1141	975	753	451	187	20	3	6	89	398	731	1099		
	CDD50	3327	0	3	37	117	347	640	807	715	476	160	24	1		
	CDD65	997	0	0	1	6	56	210	344	256	113	11	0	0		
	CDH74	10219	0	2	31	146	708	2202	3471	2253	1219	178	9	0		
	CDH80	4214	0	0	5	37	244	903	1598	909	471	47	0	0		
(14) Wind	WSAvg	10.3	9.9	10.6	11.9	13.1	11.9	10.6	8.4	7.7	9.4	10.2	10.1	9.7		
(15) Precipitation	PrecAvg	25.8	0.6	0.6	1.8	2.1	4.3	4.1	3.9	3.1	2.3	1.6	1.0	0.5		
	PrecMax	46.4	2.1	1.9	6.6	5.4	9.3	9.7	15.4	8.4	9.9	6.1	3.7	1.5		
	PrecMin	16.0	0.0	0.0	0.0	0.0	0.2	0.9	0.2	0.6	0.0	0.0	0.0	0.0		
	PrecStd	6.0	0.5	0.5	1.5	1.3	2.2	2.3	3.1	1.7	2.0	1.4	1.0	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.6	69.2	77.5	85.5	91.4	96.9	99.9	97.7	94.5	86.4	74.7	61.1		
		MCWB	45.8	49.8	54.7	58.4	64.6	70.1	72.7	71.8	67.8	62.6	53.2	45.2		
	2%	DB	54.7	59.0	72.0	78.6	86.4	92.6	95.3	92.7	90.1	79.0	67.8	54.0		
		MCWB	42.2	44.9	52.7	57.5	65.3	70.4	73.7	72.7	67.9	60.1	51.1	43.2		
	5%	DB	48.4	53.1	65.6	72.5	81.6	89.6	91.5	89.3	85.8	73.3	61.5	48.3		
		MCWB	39.2	42.7	51.3	55.7	63.1	70.3	73.2	72.8	67.2	57.4	48.3	39.9		
	10%	DB	43.4	46.1	59.5	67.7	77.2	85.5	89.3	85.2	81.4	69.0	55.4	44.0		
		MCWB	36.4	38.6	47.8	53.7	61.9	68.6	73.1	71.7	64.8	55.7	45.7	36.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.3	50.7	60.3	62.5	70.4	76.4	79.0	77.9	73.4	66.9	56.9	49.6		
		MCDB	61.0	63.9	68.7	74.3	82.7	87.3	90.3	88.7	85.5	77.7	65.6	59.8		
	2%	WB	42.7	45.7	56.4	59.9	68.3	73.5	76.8	75.9	71.0	63.3	52.8	43.4		
		MCDB	53.3	57.5	66.3	72.8	80.4	86.1	88.7	85.9	82.4	71.8	62.8	52.9		
	5%	WB	39.9	42.4	52.3	57.2	66.0	71.8	75.3	74.1	69.1	60.2	49.8	40.3		
		MCDB	47.5	51.9	63.0	70.2	77.9	84.8	87.1	84.4	80.2	69.9	60.9	48.1		
	10%	WB	37.0	39.0	48.4	54.8	63.4	70.1	73.9	72.4	67.4	57.2	46.3	37.2		
		MCDB	43.0	46.3	60.4	65.7	73.3	82.4	85.5	82.4	78.0	66.9	56.1	43.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.9	20.0	24.3	23.6	22.2	22.2	21.7	21.1	24.3	23.8	22.7	19.2		
		MCDBR	28.2	30.4	34.8	32.5	28.7	27.2	26.8	26.6	29.3	31.1	31.4	26.4		
		MCWBR	17.8	18.2	18.9	16.3	12.6	11.1	11.3	11.1	11.5	14.9	17.5	17.2		
		MCDBR	26.1	28.5	30.6	28.2	25.1	24.1	23.0	22.7	24.3	25.3	29.5	25.1		
	5% WB	MCWBR	17.1	17.7	18.0	15.3	12.5	11.6	11.5	11.3	11.4	14.3	17.2	17.1		
(40) Clear-Sky Solar Irradiance	taub		0.270	0.274	0.300	0.334	0.361	0.377	0.395	0.389	0.353	0.311	0.274	0.265		
	taud		2.506	2.512	2.466	2.387	2.387	2.381	2.367	2.370	2.424	2.522	2.542	2.519		
	Ebn at Noon		288	303	304	298	290	285	278	277	281	285	285	282		